



Pigeon Hole

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Courtesy: slides are adopted from Discrete Mathematics and Its Applications by Kenneth H. Rosen

Pigeon Hole Principle

Simple Form: If we distribute $n + 1$ balls into n bins, then there is at least a bin which contains at least two balls

General Form: If we distribute $kn + 1$ balls into n bins, then there is at least a bin which contains at least $k + 1$ balls.

Proof: For the sake of contradiction, assume each bin contains at most k ball. Then the number of balls is going to be at most nk which is a contradiction.

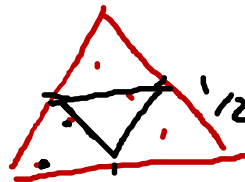
Pigeon Hole Principle

General: If we distribute m balls into n bins, then there is at least a bin which contains at least $\lceil m/n \rceil$ balls.

اثبات: زمین می‌سیم در هر خانه حداکثر $\lceil \frac{m}{n} \rceil - 1$ آپ قرار می‌گیرد.

$$\text{تعداد آپ} \leq n(\lceil \frac{m}{n} \rceil - 1) < n(\frac{m}{n} + 1 - 1) = m.$$

مثال: از هر دو نقطه درون یک مثلث متساوی‌الساق به ضلع ۱، فاصله که حداکثر ۱/۲ است.



Some Problems

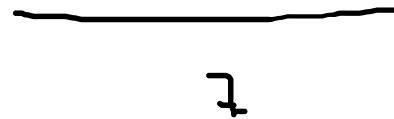
Problem: There are two persons out of 8 persons whose birthday happened in the same day of the week.

Solution:

Balls: 8 persons

Bins: 7 days of the week

Assigning balls into bins: If a person was born for instance on Sunday, it goes to the bin whose label is Sunday.



Some Problems

Problem: Assume $A \subseteq \{1, \dots, 2n\}$, $|A| = \overline{n + 1}$. Show that there are two numbers $\underline{a}, \underline{b} \in A$ s.t. $\underline{a} | \underline{b}$

Solution:

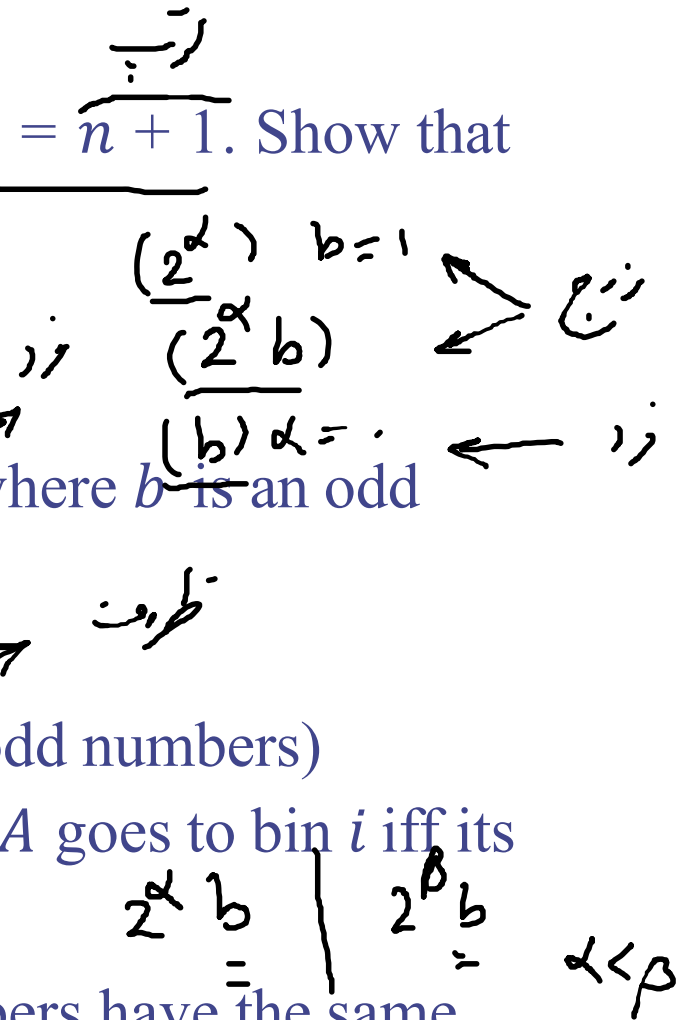
Each number n can be written as $2^\alpha \underline{b}$ where $\underline{b}$ is an odd number (the odd part of n).

Balls: elements of A .

Bins: elements of $\{1, 3, \dots, 2n - 1\}$ (odd numbers)

Assigning balls into bins: an element of A goes to bin i iff its odd part is i .

#balls = $n + 1$, #bins = n , then two numbers have the same odd part, So, one is divisible by the other.



Some Problems

Problem: Assume $A \subseteq \{1, \dots, 2n\}$, $|A| = n + 1$. Show that there are two numbers $a, b \in A$ s.t. $(a, b) = 1$

Solution:

Balls: elements of A .

Bins: the pairs $(2i - 1, 2i)$ ($i = 1, \dots, n$)

Assigning balls into bins: an element of A goes to bin $(2i - 1, 2i)$ iff it is equal to either $2i - 1$ or $2i$. #balls = $n + 1$, #bins = n , then two numbers are in the same bin. These two numbers are relatively prime as they are consecutive.

$$(1, 2) (3, 4) (5, 6) \dots (2n-1, 2n)$$

$\underbrace{\hspace{10em}}_{i \leq n}$

$$\begin{aligned} d &| n \\ d &| n+1 \end{aligned}$$

$$\begin{aligned} d &| (n+1) - n \\ \Rightarrow d &= 1 \end{aligned}$$

Some Problems

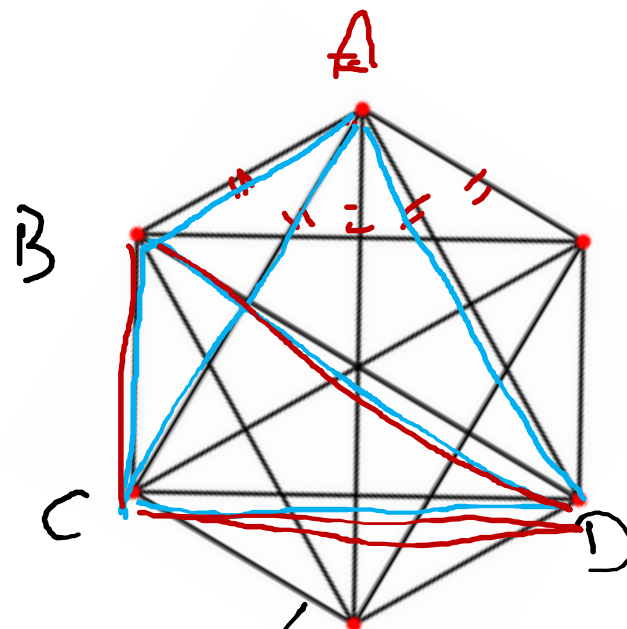
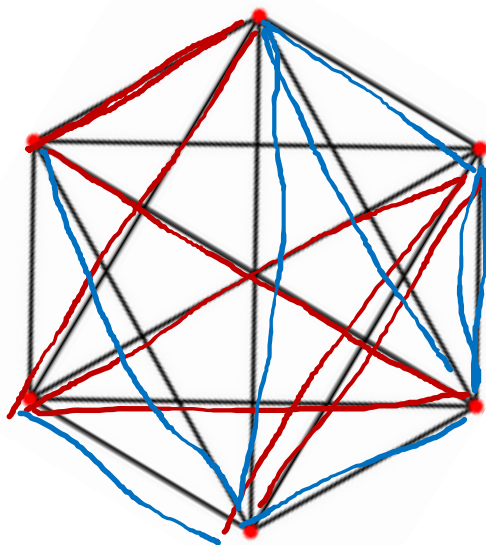
Problem: Consider 6 distinct points and all segments whose endpoints are these 6 points. We color the segment with either red or blue. Prove that there is a triangle whose edges have the same color.

Solution:

Consider a point A. It has 5 segments incident to it. At least three of them have the same color, say red.

Call the other endpoints of these three red segments B, C and D. If one of segments BC, CD and DB is red, we have a red triangle. Otherwise all three must be blue.

Then we have a blue triangle. In both cases, we have a triangle whose edges have the same color.



۵-۲ - ۲ رنگ
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 ۳-۳ رنگ

Some Problems

Problem: We have 9 numbers of form $2^\alpha 3^\beta 5^\gamma$. Show that there are two numbers whose product is a square number.

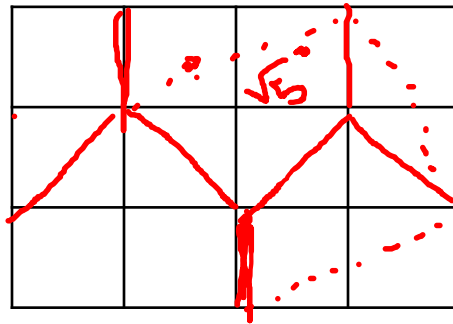
Solution:

Balls: 9 numbers

Bins: 8 triples (a, b, c) where $a, b, c \in \{0, 1\}$

Assigning balls into bins: ball $2^\alpha 3^\beta 5^\gamma$ goes into bin (a, b, c) iff $\alpha \equiv a \pmod{2}, \beta \equiv b \pmod{2}, \gamma \equiv c \pmod{2}$,
#balls= 9 and #bin= 8. Then, two numbers are in the same bin.
So, the product of these two numbers is a square number.

مثال : به ازای ۶ نقطه داخل مستطیل 2×4
 حداقل درجه وجود دارد که به این
 کمتر از ۵ است.



۶ نقطه، ۵ نامنه

Some Problems

Problem: Ali read his lecture notes 30 hours during the last 20 days to get prepared for the final exam. At the i -th day of his preparation, he spent a_i hours where a_i is a natural number. Prove that there were some consecutive days s.t. the total time he spent on these days is exactly 9 hours.

Solution:

Let $x_i = \sum_{j=1}^i a_j$. Clear that $x_i \in \{1, 2, \dots, 30\}$.

let $y_i = x_i + 9$. Clear that $y_i \in \{10, 11, \dots, 39\}$. So, $x_i, y_i \in \{1, 2, \dots, 39\}$. Also clear that

$x_1 < x_2 < \dots < x_{20}$ and $y_1 < y_2 < \dots < y_{20}$.

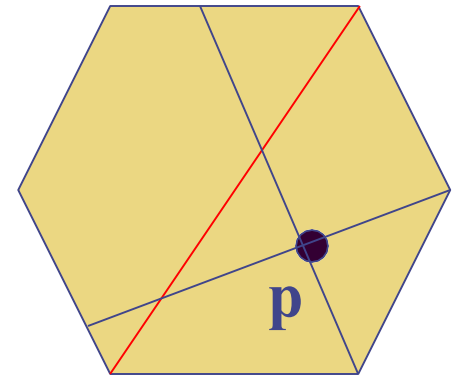
We have 40 numbers; each belonging to $\{1, 2, \dots, 39\}$. Then two of them must be equal. So, $x_i = y_j$ for some i and j . So $x_i - x_j = 9$.

Some Problems

Problem: Assume we have a convex polygon with $2n$ vertices and a point p inside it which is not on any diagonal of the polygon. For each vertex A , consider the line passing through A and p . This line intersects an edge of the polygon. Prove that there is an edge of the polygon not being intersected by these lines.

Solution:

- $A_1A_2 \dots A_{2n}$ are the vertices.
- p lies in one side of A_1A_{n+1} .
- Assume p is inside $A_1A_2 \dots A_nA_{n+1}$.
- Only lines $A_2p, A_3p, \dots, A_np$ ($n-1$ lines) may intersect edges $A_{n+1}A_{n+2}, A_{n+2}A_{n+3}, \dots, A_{2n}A_1$ (n edges).
- Then one edge must be free of intersection



Some Problems

Problem: Suppose we have a sequence of distinct numbers of length $n^2 + 1$. Prove that there is a subsequence of length $n + 1$ which is either decreasing or increasing.

Solution:

- $\langle a_1, \dots, a_{n^2+1} \rangle$ is the sequence
- Assign a_k a pair (i_k, d_k) where i_k (d_k) is the length of the longest increasing (decreasing) subsequence starting (ending) at a_k
- If one of i_k s and d_k s is greater than n , we are done
- Otherwise, $n^2 + 1$ pairs (i_k, d_k) are of form (a, b) where $1 \leq a, b \leq n$. So, there are k and l s.t. $(i_k, d_k) = (i_l, d_l)$
- Assume $k < l$. if $a_k < a_l$ ($a_k > a_l$), then $i_k(d_l)$ must be at least $i_l + 1$ ($d_k + 1$)

